

UC10 — CANONICAL DATA KNOWLEDGE GUIDE

Data-First, Feature-First, Detector-Second

SECTION 1 — WHAT DATA DO WE HAVE?

We rely on three core synthesized CMS datasets. The following metrics reflect the exact audited state of our data sample:

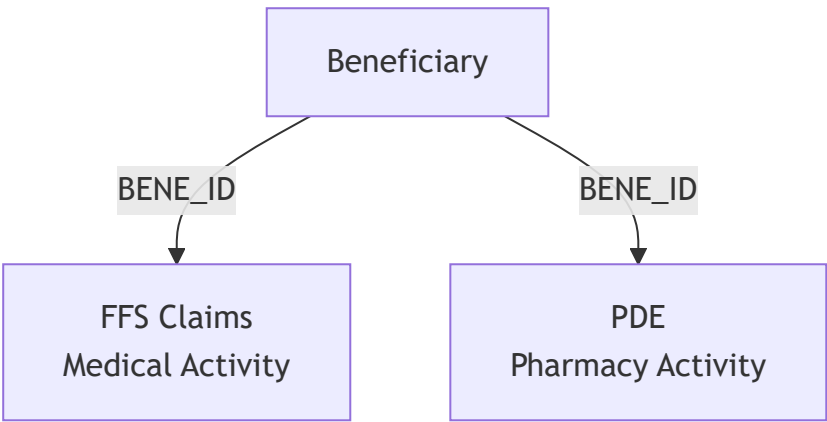
Dataset	What it represents	Grain	Key	Important date	Why UC10 uses it
Beneficiary	member/reference view	1 row per beneficiary	BENE_ID	(Reference)	Master population list
FFS Claims	medical activity	claim-line grain	CLM_ID / CLM_LINE_NUM	CLM_FROM_DT / NCH_WKLY_PROC_DT	Baseline medical volume
PDE	pharmacy activity	event-level grain	PDE_ID	SRVC_DT	Baseline pharmacy volume

Audited Dataset Facts:

- **Beneficiary:** 5,975 rows, 123 columns.
- **FFS Claims:** 1,121,004 rows, 96 columns. Maps to 90,705 unique claims across 7,971 unique beneficiaries.
- **PDE:** 515,520 rows, 36 columns. Maps to 515,520 unique events across 7,403 unique beneficiaries.

SECTION 2 — HOW ARE THE DATASETS RELATED?

BENE_ID acts as the primary relational key bridging medical and pharmacy activity.



Audited Relationship Findings:

- Total Beneficiaries in Master: 5,975
- Intersection (Claims + PDE): 6,850 beneficiaries have both medical and pharmacy events.
- **Orphan Claims:** 2,436 unique beneficiaries in Claims do not exist in Beneficiary.
- **Orphan PDEs:** 2,018 unique beneficiaries in PDE do not exist in Beneficiary.

What this means for UC10:

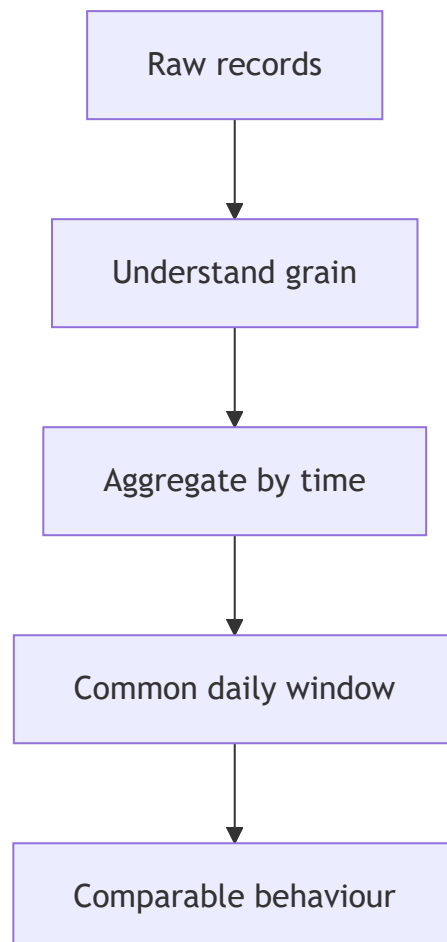
We do not possess perfect referential integrity. These orphan records are not a mistake to be silently cleaned; they serve as active Data Quality (DQ) evidence that the anomaly detectors monitor.

SECTION 3 — WHY GRAIN MATTERS

Understanding the data "grain" is critical:

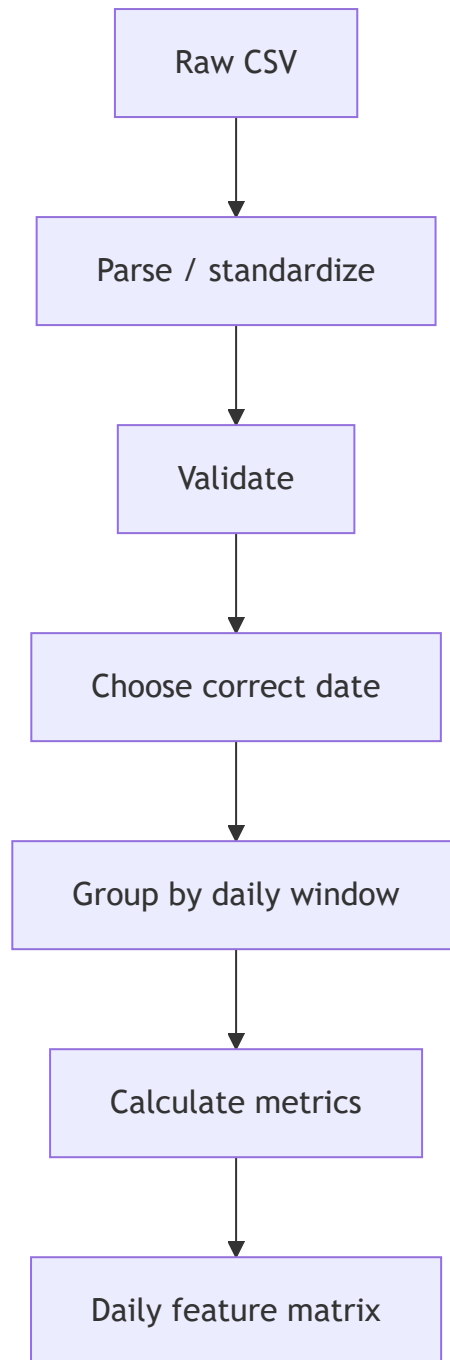
- **Beneficiary:** 1 row = 1 beneficiary
- **Claims:** 1 row = 1 claim line
- **PDE:** 1 row = 1 prescription event

Because a single complex medical claim can generate hundreds of line items, comparing raw claim rows directly to single-event pharmacy rows is meaningless. We must aggregate these varying granularities into a daily window before the data can be analyzed.



SECTION 4 — FROM RAW RECORDS TO DAILY DATA

The machine learning model does not directly consume millions of raw healthcare records. It consumes a compact daily representation of behaviour.



SECTION 5 — THE FOUR DATA ASPECTS

A. DATA QUALITY

Question: "Is the data valid?"

Features: null_rate, duplicate_rate, claims_null_rate, cross_dataset_mismatch_rate, dq_violation_rate

B. HEALTHCARE VOLUME / BEHAVIOUR

Question: "Is healthcare activity behaving normally?"

Features: claim_count, pde_count, unique_beneficiary_count, unique_provider_count, claim_pde_ratio, volume_change_pct

C. DISTRIBUTION

Question: "Have the characteristics of the data changed?"

Features: median_days_supply, median_rx_cost, median_claim_amount

D. OPERATIONAL TELEMETRY

Question: "Is the pipeline processing normally?"

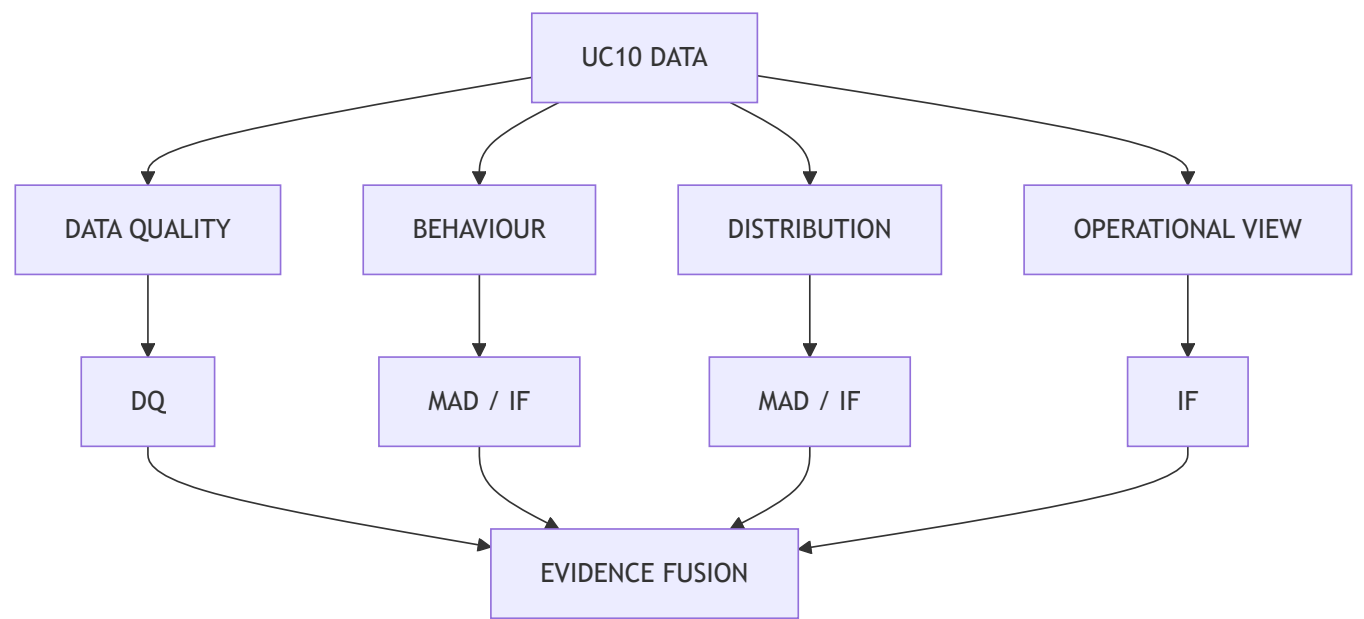
Features: processing_duration, throughput, failure_rate, backlog

[PROTOTYPE-SIMULATED]

SECTION 6 — FEATURE DICTIONARY

Feature	Built from	What it tells us	Aspect	Detector
claim_count	FFS Claims	daily medical activity volume	healthcare behaviour	MAD / IF
pde_count	PDE	daily pharmacy activity volume	healthcare behaviour	MAD / IF
unique_beneficiary_count	Claims + PDE	unique patients active	healthcare behaviour	IF
unique_provider_count	Claims + PDE	unique billing providers	healthcare behaviour	IF
claim_pde_ratio	Claims + PDE	relationship between medical and pharmacy activity	cross-dataset behaviour	IF
volume_change_pct	Derived	week-over-week or day-over-day volatility	healthcare behaviour	MAD
median_days_supply	PDE	shift in prescription duration	distribution	MAD / IF
median_rx_cost	PDE	shift in average drug prices	distribution	MAD / IF
median_claim_amount	FFS Claims	shift in average medical billing	distribution	MAD / IF
null_rate	DQ checks	missing critical identifiers across files	data quality	DQ
duplicate_rate	DQ checks	repeating identical primary keys	data quality	DQ
claims_null_rate	DQ checks	missing diagnosis or provider codes	data quality	DQ
cross_dataset_mismatch_rate	Claims + PDE	orphan records missing BENE_ID references	data quality	DQ
dq_violation_rate	DQ checks	overall data rule failures	data quality	DQ
processing_duration	Pipeline metrics	batch processing execution time	operational	IF / SLA
throughput	Pipeline metrics	processed records per second	operational	IF / SLA
failure_rate	Pipeline metrics	failed processing tasks	operational	IF / SLA
backlog	Pipeline metrics	volume of pending work	operational	IF / SLA

SECTION 7 — FEATURE → ASPECT → DETECTOR



Data Aspect	Features	Main Question	Primary Detector
Data Quality	null_rate, duplicate_rate, etc.	Is data invalid?	DQ
Volume	claim_count, pde_count	Is volume unusual?	MAD
Cross-data behaviour	claim_pde_ratio	Are streams behaving differently?	IF
Distribution	median_rx_cost, etc.	Has the distribution shifted?	MAD / IF
Operational	throughput, backlog, etc.	Is processing degrading?	IF / SLA

SECTION 8 — WHY EACH DETECTOR EXISTS

- **DQ:** "Something is explicitly wrong."
Example: Missing BENE_ID → DQ
- **MAD:** "One measurement is unusually different from its recent normal."
Example: 40% volume drop → MAD
- **Isolation Forest:** "Several measurements look unusual together."
Example: Volume slightly abnormal + throughput decreasing + backlog increasing → Isolation Forest

SECTION 9 — WHAT IS REAL AND WHAT IS SIMULATED?

The current CMS datasets describe healthcare activity, not live pipeline execution. Operational telemetry is therefore simulated for the prototype.

REAL DATA

- claim_count
- pde_count
- beneficiary counts
- provider counts
- median costs
- median days supply
- DQ metrics
- cross-data relationships

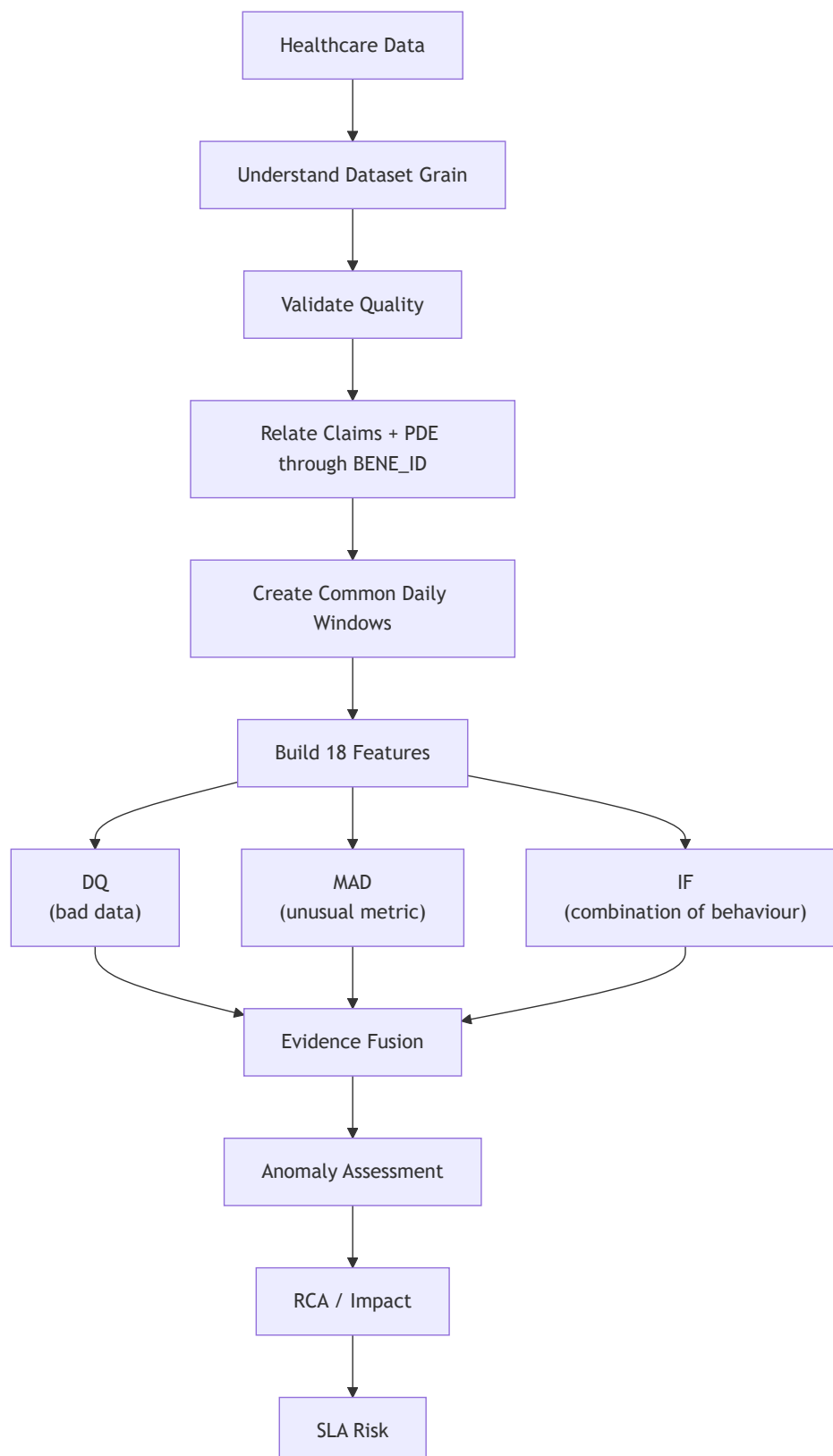
SIMULATED OPERATIONAL TELEMETRY

- processing_duration
- throughput
- failure_rate
- backlog
- SLA-related metrics

SECTION 10 — ANOMALY COVERAGE

Anomaly	Evidence	Feature	Detector
Missing ID	null identifier	null_rate	DQ
Duplicate	repeated record/key	duplicate_rate	DQ
Invalid date	invalid date	dq_violation_rate	DQ
Referential mismatch	orphan/reference mismatch	cross_dataset_mismatch_rate	DQ
Volume drop	sudden volume change	claim_count / pde_count	MAD
Multivariate degradation	several abnormal metrics	behavioural/operational features	IF

SECTION 11 — COMPLETE DATA STORY



SECTION 12 — WHAT THIS DATA CAN AND CANNOT PROVE

CAN SUPPORT:

- healthcare activity monitoring
- data quality monitoring
- cross-dataset consistency analysis
- temporal volume anomaly detection
- multivariate behavioural anomaly detection
- prototype SLA modelling using simulated telemetry

CANNOT DIRECTLY PROVIDE:

- real production pipeline execution telemetry
- real processing duration
- real queue backlog
- real production SLA deadlines
- real prior authorization workflow data
- real authentication/authorization transactions

SECTION 13 — FINAL TAKEAWAY

The UC10 data foundation converts heterogeneous healthcare records into a common daily behavioural view. The resulting features provide different types of evidence: data-quality evidence, volume and behaviour evidence, distribution evidence, and prototype operational evidence. Each evidence type is sent to the detector best suited to identify that type of anomaly.

DATA FOUNDATION STATUS: READY TO FREEZE